[Assignment-3]

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section : 28

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Ans to the question no 1

a) Perspective projection is considered more realistic for simulating human vision because it mimics the way human eyes perceive objects, where objects appear smaller as they move farther away. This creates a sense of depth and distance, which is essential for immersive experience in 3D graphics, such as video games or simulations. For example, in a 3d game, a character moving away from the viewer will gradually decrease in size, enhancing realism.

Parallel projection is preferred in engineering and architectural drawing because it preserves exact dimension and angles, which is critical for technical accuracy. In parallel projection, lines remain parallel and do not converge, allowing measurements to be taken directly from the drawing. For instance, in a blue print of a building, parallel projection ensures that all walls and features are drawn to scale without distortion, facilitating construction and design.

Given, $P(10, 20, -40)$

b)

$x' = x + z\cos\beta$ $\cancel{10+}$

$y' = y + z\sin\beta$

$z' = 0$

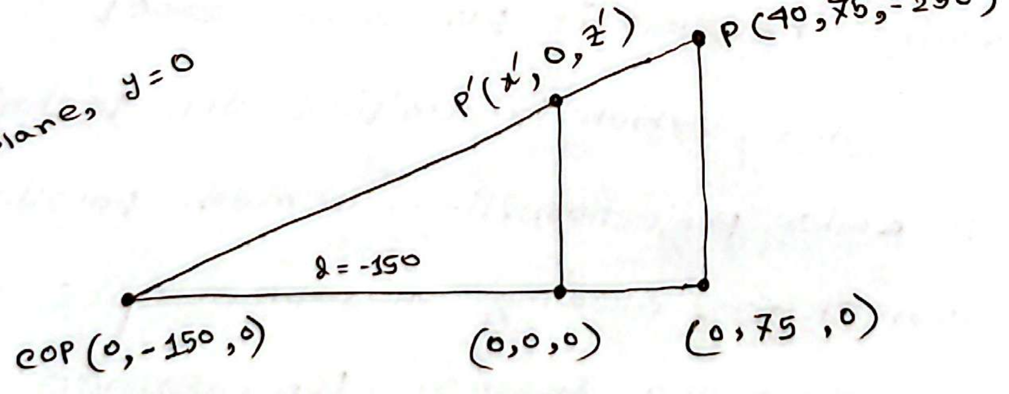
$\beta = 30°$

$$\therefore P' = \begin{bmatrix} 1 & 0 & \cos 30° & 0 \\ 0 & 1 & \sin 30° & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -24.64 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore P'(-24.64, 0, 0)$ (Ans)

c) xz plane, $y = 0$



$$\therefore P' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \frac{1}{150} & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 40 \\ 75 \\ -250 \\ 1 \end{bmatrix} = \begin{bmatrix} 40 \\ 0 \\ -250 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 26.67 \\ 0 \\ -166.67 \\ 1 \end{bmatrix}$$

$\therefore P'(26.67, 0, -166.67)$

(Ans)

Ans to the question no 2

a) The projection described is Cavalier Projection.

Two defining properties are:

1. The projectors are parallel and make an angle with the projection plane.

2. Lines perpendicular to the projection plane are projected at true length (no forshortening).

If the axis were reduced to half its actual length, it would become Cabinet projection. The visual appearance would show less distortion and appear more natural due to the forshortening effect, making the depth look more proportional.

b)

Point A (200, 150, 360)

Point B (600, 130, 360)

Panel → marker distance = 360 - 120 = 240

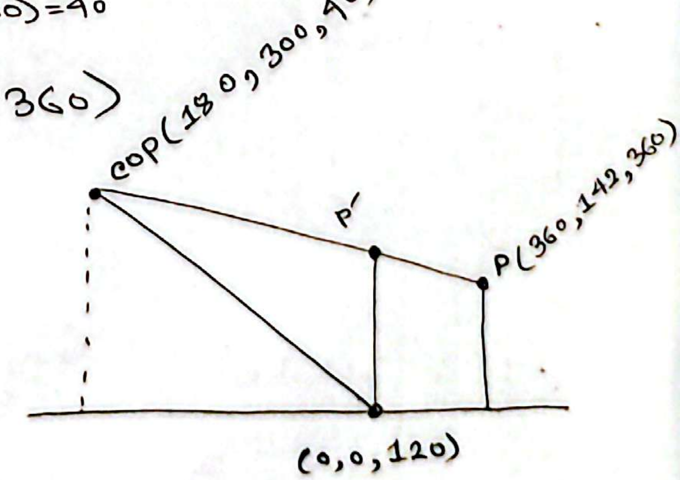
Eye(COP) → panel " $= \frac{240}{3} = 80$

$COP_z = (120 - 80) = 40$

Vector, $\bar{V} = (600, 130, 360) - (200, 150, 360)$

$= (400, -20, 0)$

40% of $\bar{V} = (160, -8, 0)$



Thus,

The marker spot, P = A + (160, -8, 0)

= (360, 142, 360)

$$P' = \begin{bmatrix} 1 & 0 & \frac{-180}{-80} & \left(120 \times \frac{180}{-80}\right) \\ 0 & 1 & \frac{-300}{-80} & \left(120 \times \frac{300}{-80}\right) \\ 0 & 0 & \frac{-120}{-80} & \left(120 + \frac{(120)^2}{-80}\right) \\ 0 & 0 & \frac{-1}{-80} & \left(1 + \frac{120}{-80}\right) \end{bmatrix} \times \begin{bmatrix} 360 \\ 142 \\ 360 \\ 1 \end{bmatrix}$$

$$= \cancel{\begin{bmatrix} -720 \\ -1658 \\ -600 \\ -5 \end{bmatrix}} = \begin{bmatrix} 900 \\ 1042 \\ 480 \\ 4 \end{bmatrix}$$

$q_x = +180$

$q_y = +300$

$q_z = COP_z - z_p$

$= 40 - 120$

$= -80$

$$= \cancel{\begin{bmatrix} -720/-5 \\ -1658/-5 \\ -600/-5 \\ -5/-5 \end{bmatrix}} = \begin{bmatrix} 900/4 \\ 1042/4 \\ 480/4 \\ 4/4 \end{bmatrix}$$

$$= \cancel{\begin{bmatrix} 144 \\ 331.6 \\ 120 \\ 1 \end{bmatrix}} = \begin{bmatrix} 225 \\ 260.5 \\ 120 \\ 1 \end{bmatrix}$$

~~∴ P′(144, 331.6, 120)~~

~~(Ans)~~

∴ P′(225, 260.5, 120)

(Ans)